YUANFEI HU

NeuroAI · Applied Mathematics · Machine Learning · Data Science

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EDUCATIONAL BACKGROUND

National University of Singapore — **Singapore**
M.Sc. in Biomedical Informatics, Yong Loo Lin School of Medicine — *2026.08–2027.05 (expected)*

Duke Kunshan University & Duke University — **China / United States**
Dual-Degree Program, Class of 2026
B.S. in Applied Mathematics & Computational Science (Math Track) — *2022.08–2026.05*
B.A. in Interdisciplinary Studies — Applied Mathematics & Computational Science — *2024.06–2024.12*

- **Relevant Coursework:** Mathematical Basis of Machine Learning, Probability & Statistics, Linear Algebra, Dynamical Systems, Computational Neuroscience, Cognitive Neuroscience, and Partial Differential Equations.
- **Honors:** Chancellor's Scholarship (one recipient per cohort; full scholarship); Dean's List with Distinction (Top 10%); 1st in Sichuan Science Evaluation.

RESEARCH EXPERIENCE

Geometry-Informed SE(2) Consistency Monitoring for Robotic Fault Detection — **Remote**
Independent Research Collaborator & Co-first Author — *2026.07–Present*
Manuscript submitted to *Robotics and Autonomous Systems*; co-first author, listed first.
Reproducibility package: DOI 10.5281/zenodo.22208341

- Co-first-authored a submitted robotics manuscript on geometry-informed fault monitoring under SE(2) coordinate-frame transformations, with a public reproducibility archive containing code, frozen protocols, checkpoints, raw outputs, and figures.
- Developed a lightweight neural controller and geometry-consistency detector that combines equivariance residuals, kinematic deviations, and command–observation mismatches; calibrated the detector using healthy trajectories only.
- Designed locked validation, coordinate-frame/OOD stress tests, multi-seed regularization ablations, fault-family controls, and recovery-control ablations across eight simulated robotic fault types.
- Achieved 0.8568 macro AUROC on the untouched locked q50 validation partition; documented multi-seed uncertainty and non-significant effects transparently rather than optimizing post hoc.

Sleep Replay on Continuous Memory Manifolds — **Kunshan / Singapore**
Independent Researcher — Signature Work Distinction — *2025.04–Present*
GitHub: sleep-replay-continuous-manifold

- Investigated a gap in standard memory-consolidation theories: sleep may not simply strengthen memories, but can either blend similar experiences together or make them easier to distinguish depending on replay timing and neural overlap.
- Built a label-free computational model and designed a 4,032-row simulation study with matched controls to isolate these effects; identified a bounded regime in which replay separated previously confused memories by 0.0517 radians while also improving precision and reducing recall error by 0.00710 radians.
- Stress-tested the finding through parameter sweeps, a locked replication in 48 simulated subjects, a biologically constrained excitatory/inhibitory implementation, and model-recovery experiments under sparse and noisy conditions, demonstrating rigorous experimental design and reproducible computational research.
- Reanalyzed two public targeted-memory-reactivation datasets using fixed-effects modeling, held-out prediction, bootstrap inference, and permutation tests; distinguished predictions supported by existing behavioral data from biological claims that still require item-level physiological validation.

Jianfeng's Lab — **Durham, USA**
Research Assistant — Professor Jianfeng Lu — *2024.07–2024.12*

- Developed and tuned attractor-network models of memory retrieval, applying Dynamic Mean-Field Theory to quantify transitions among recall, drift, and memory loss and improve dynamical stability and biological plausibility.
- Reproduced results from *Physical Review X* 13, 011009 in MATLAB across 300+ simulations, measuring retrieval errors and validating predicted network regimes over parameter sweeps.
- Produced reusable code and technical documentation for future lab replication; received a $1,000 student research award.

Nonlinear Dynamics & Forecasting Project **Kunshan, China**
Project Lead — Outstanding Student Honor *2025.07–2025.09*

- Processed 150+ multi-trial CO_2 time series and constructed analysis-ready datasets for indoor air-quality forecasting and nonlinear dynamical modeling.
- Built and compared nonlinear ODE, LSTM, and tree-based models, improving predictive accuracy by 17–24% over ODE baselines and communicating model trade-offs through quantitative visualizations.
- Led a five-member team across experimental design, data processing, model evaluation, and presentation of results for environmental-monitoring applications.

DKU iGEM 2024 — Silver Medal **Kunshan, China**
Modeling Team Leader *2023.10–2024.09*

- Led computational modeling within a 12-member interdisciplinary team, translating wet-lab measurements and approximately 20 research papers into mathematical assumptions, parameter choices, and validation strategies.
- Developed multiscale diffusion-gradient and degradation simulations that explained more than 60% of observed experimental variance for a synthetic-biology platform designed to support the vaginal microbiome.
- Integrated modeling results with the experimental workflow and built web-based visualizations to communicate methods and findings to technical and public audiences.

SELECTED PROJECT EXPERIENCE

DKU Dii Project: Shape of Pain **Kunshan, China**
Project Founder — Supervisor Shan Wang *2023.08–2024.05*

- Led an interdisciplinary project combining capsaicin-induced blood-flow heatmaps with PDE-based diffusion models to quantify and visualize physiological pain responses.
- Developed a multimodal representation framework connecting pain measurements with visual, auditory, and tactile descriptors; secured top-tier seed funding for continued development.

LEADERSHIP EXPERIENCE

DKU Travel Club **Kunshan, China**
President; Secretary & Design Section Head *2023.09–Present*

- Led a 200+ member organization across finance, logistics, media, and event operations; delivered five 30–40-person excursions with 98%+ participant satisfaction.
- Founded JourneyConnect, won first place in a campus pitch competition, and secured RMB 25,000 in seed funding for a cross-campus travel-networking platform.

DKU Residence Life **Kunshan, China**
Resident Assistant *2025.05–Present*

- Oversaw residential community of 35 students, organized 9+ events in 25Fall semester while ensuring safety compliance and community standards;
- Provided crisis management for 3+ emergencies, 2+ conflicts, and 1 mental health situation, cultivated supportive environment through mentorship, programming, and proactive community engagement initiatives.

ADDITIONAL INFORMATION

- **Programming & Data:** Python, SQL, C++, MATLAB; NumPy, SciPy, pandas, scikit-learn, and Matplotlib.
- **Machine Learning:** PyTorch, TensorFlow, JAX; neural networks, geometric/equivariant learning, anomaly detection, time-series forecasting, supervised and unsupervised learning.
- **Statistics & Modeling:** Bootstrap and permutation inference, cross-validation, fixed-effects models, dynamical systems, attractor networks, experimental design, and model evaluation.
- **Engineering Tools:** Git, GitHub, Jupyter, LaTeX; Linux and Docker (familiar); reproducible experiments and data pipelines.
- **Languages:** Mandarin (Native), English (Fluent; TOEFL 117), Spanish (Conversational).
- **Awards:** National First-Class Athlete; Royal Academy of Dance A2 Distinction; DKU Athletic Champion (2022, 2023); DKU Global Leadership Program Distinction.